

Automated defect detection in PV panels using electroluminescence imaging and optimized YOLO models to ensure grid compliance and reliability.

Energy × Regulatory compliance and reporting × Computer vision · OS146 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
48 is the world moving	50 can we play, can we win	MEDIUM 0 named in the evidence	€22.6m observed evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

Automated defect detection in PV panels using electroluminescence imaging and optimized YOLO models helps solar plant operators ensure grid compliance and reliability. This opportunity targets energy companies and asset owners who need to maintain optimal energy production and meet regulatory reporting requirements. The need is urgent because modern grids require secure, reliable integration of renewable energy, and manual inspection is no longer sufficient.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€27.5m €6.1m – €348m	€22.6m €5.1m – €287m	€272k €61k – €3.4m	observed
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Computer vision	6.8 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TIR	2025
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

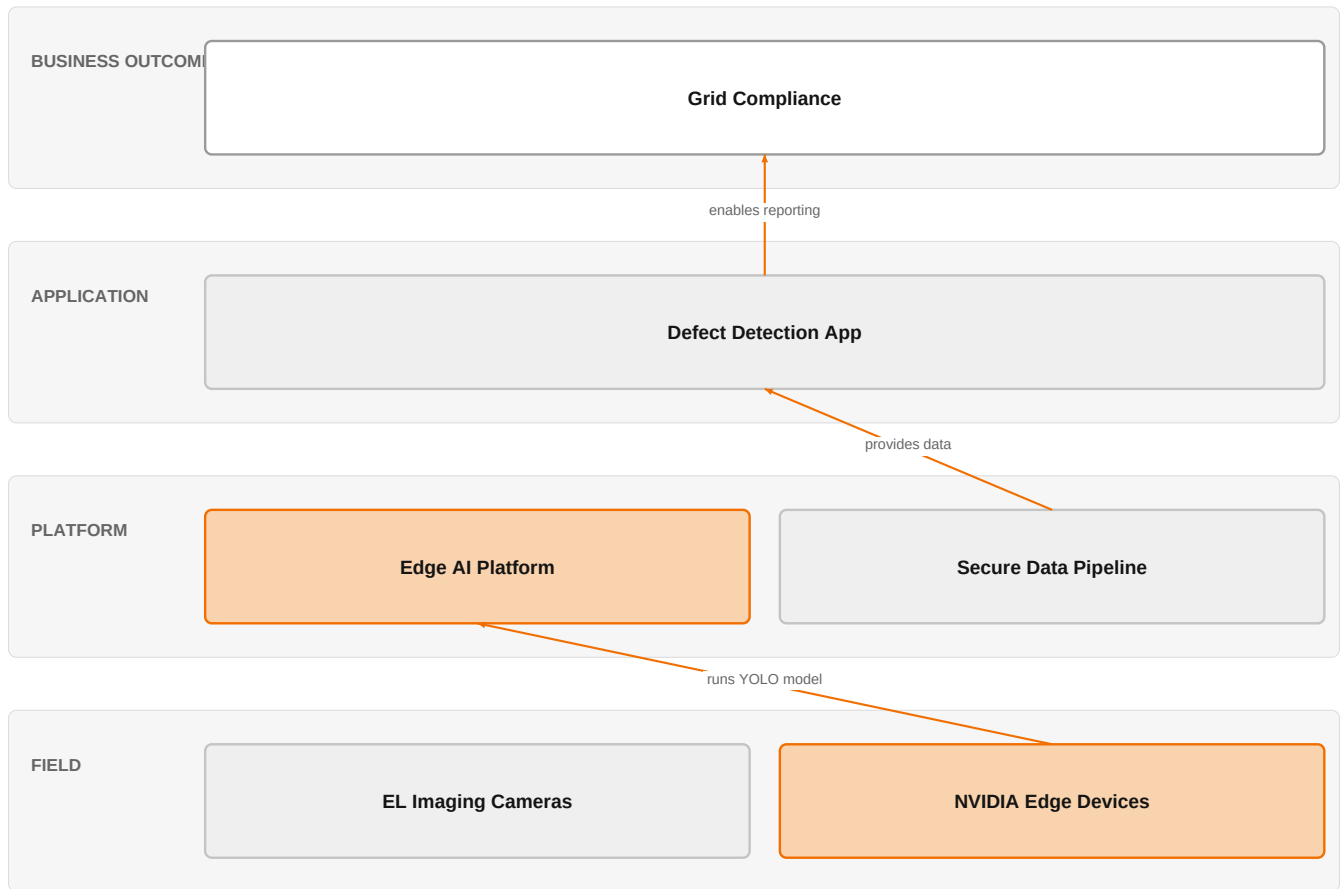
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

PV Defect Detection Solution



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This solution combines edge AI with secure data handling to automate PV defect detection and support grid compliance.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The rapid replacement of synchronous generators with inverter-based resources has reduced system inertia, making grid stability a priority and driving the need for reliable PV integration (SIG-B81FDE60EE21). Deep learning models like PV-YOLOv12n now automate defect detection in electroluminescence images, which is essential for maintaining optimal energy production and reliability (SIG-AA5215E67485). AI-integrated autonomous robotics combine real-time monitoring and predictive analytics to address performance degradation from dust and heat (SIG-39681DB73044). These advances make automated, vision-based inspection practical and necessary for grid compliance.

Sources: SIG-B81FDE60EE21 cordis.europa.eu 2026-05-01; SIG-AA5215E67485 openalex.org 2025-09-26; SIG-39681DB73044 openalex.org 2025-09-01

Who buys, and why now

The buyer is typically the head of asset management or operations at a utility-scale solar plant or an energy company's renewable division. The pain is felt by field maintenance teams who must manually inspect thousands of panels, a tedious and error-prone process. The trigger is often a grid compliance audit or a performance degradation event that forces the operator to demonstrate panel health and reliability.

Why it is hot now — every claim bound to a dated source

- SUNVIBE aims to ensure secure, reliable integration of renewable energy into Europe's grids. [SIG-B81FDE60EE21]
- PV-YOLOv12n automates defect detection in electroluminescence images of PV panels, which is essential for maintaining optimal energy production and reliability. [SIG-AA5215E67485]

- Optimized YOLO-based models are being developed for defect detection in electroluminescence images of PV panels to ensure reliability and optimal energy production. [SIG-AA5215E67485]
- AI-integrated autonomous robotics combine real-time monitoring and predictive analytics for solar panel performance, addressing dust and heat degradation. [SIG-39681DB73044]
- Deep learning models automate defect detection in PV panels, improving reliability and energy production. [SIG-AA5215E67485]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble an end-to-end solution using its IoT experts to deploy edge devices with cameras for electroluminescence imaging, and Orange Group researchers to fine-tune an optimized YOLO model for defect detection. The solution would run on NVIDIA edge hardware, with data processed locally for low latency. Orange Cyberdefense managed services would secure the data pipeline, and the entire system would be certified to ISO 27001 and SecNumCloud, ensuring sovereign and secure operations.

Why Orange can win

Orange can win by combining its IoT expertise, research capability, and cybersecurity managed services with sovereign certifications (ISO 27001, SecNumCloud). The partnership with NVIDIA provides access to optimized edge hardware. This combination of secure, certified infrastructure and AI expertise is difficult for pure-play competitors to match, especially for customers with strict data sovereignty requirements.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 5.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Bosch	Industrial / OT specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
Dedrone by Axon	Security specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
HENSOLDT	Security specialist	sells Computer vision	structural	—
Honeywell	Industrial / OT specialist	sells Computer vision; active in Energy; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—
Telefonica Tech	Telecom operator peer	active in Energy	structural	—
Schneider Electric	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently inspect PV panels for defects, and what is the manual inspection frequency?
- 2 Have you experienced any grid compliance issues or performance degradation due to undetected panel defects?
- 3 What is your data sovereignty requirement for operational data collected from solar sites?
- 4 Do you have existing edge computing infrastructure at your solar plants, or would you need to deploy new hardware?
- 5 What is your current process for reporting panel health to grid operators or regulators?

Objections you will meet

They will say	You can answer
We already have a manual inspection process that works.	Manual inspection is indeed a valid baseline, but it is tedious and error-prone, and it cannot scale to the frequency needed for grid compliance. Automated detection can catch defects earlier, reducing downtime and improving reliability.
This technology is not proven at scale.	It is true that large-scale deployments are still emerging, but the research shows significant promise, and our approach leverages proven edge AI and cybersecurity practices. We can start with a pilot to demonstrate value on a subset of panels.
We are concerned about the cost of deploying new cameras and edge devices.	The initial investment is real, but it must be weighed against the cost of undetected defects, which can lead to energy losses and compliance penalties. Our solution is designed to be modular, so you can phase the deployment.

Risks and unknowns

The main risk is that the technology is still in research phase, with no large-scale commercial deployments cited. The customer may be skeptical about the maturity of YOLO-based detection in real-world conditions. Additionally, the cost of deploying electroluminescence imaging at scale may be prohibitive for some operators. It is unknown how quickly grid regulations will mandate such inspection, and whether the customer's existing infrastructure can support the required edge computing.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a solar energy client that positions Orange's ISO 27001 and SecNumCloud certifications alongside NVIDIA's acceleration to differentiate a secure, high-performance defect-detection solution, and gather evidence from Orange Group researchers on the accuracy of optimized YOLO models.
Sales	In a conversation with a European solar farm operator's operations director, open with: 'We are tracking how AI-based defect detection on PV panels can help you meet grid compliance and cut downtime — could we explore how Orange's managed security and IoT expertise could support that?'
Strategist / Innovator	Deep-dive into the PV-YOLOv12n research to assess its maturity and the feasibility of building a defect-detection service on Orange's IoT and research capabilities, and map the regulatory drivers for grid compliance in Europe to decide whether to invest in a prototype next quarter.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-B81FDE60EE21	2026-05-01	cordis.europa.eu	1	Smart Upgraded iNverter architectures for Versatile and flexible Integration of PV-Based systEms
SIG-1C6BA6D836AF	2025-12-17	openalex.org	1	FFT-Guided Multi-Window USAD with DTW–Isolation Forest for Reliable Anomaly Detection in Indust...
SIG-92E2E10DA219	2025-12-12	openalex.org	1	A real-time surface defect detection model based on adaptive feature information selection and ...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-E9AF6D858220	2025-10-21	openalex.org	1	Review on Application of Machine Vision-Based Intelligent Algorithms in Gear Defect Detection
SIG-AA5215E67485	2025-09-26	openalex.org	1	Optimized YOLO based model for photovoltaic defect detection in electroluminescence images
SIG-17D5A7F00D3D	2025-09-25	openalex.org	1	Vision transformer and Mamba-attention fusion for high-precision PCB defect detection
SIG-66E8F1ED640C	2025-09-18	openalex.org	1	Textile Defect Detection Using Artificial Intelligence and Computer Vision—A Preliminary Deep L...
SIG-39681DB73044	2025-09-01	openalex.org	1	AI-Integrated autonomous robotics for solar panel cleaning and predictive maintenance using dro...
SIG-58174080A800	2025-08-28	openalex.org	1	Enhanced PCB defect detection via HSA-RTDETR on RT-DETR
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomaly Detection in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:23+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.